Discussion Section 1
STAT 132 , Winter 2025

→ common distribution. pdf

1. Suppose that the prior distribution of some parameter θ is a beta distribution for which the mean is $1/3$ and the variance is $1/45$. Determine the prior pdf of θ.
2. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. After randomly sampling 3 students, you found that they studied 2, 2.5, and 3 hours respectively. Write out the likelihood function for λ, and then draw a sketch of this function.
3. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. Your friend suggests a uniform distribution over the interval $[0, 10]$ for λ. Is this a reasonable prior distribution for this problem? Why or why not?
4. Suppose that the proportion p of defective items in a large manufactured lot is known to be either 0.1 or 0.2. Come up with a reasonable prior distribution for this problem, and then write out $\pi(p)$.
5. Suppose that the proportion p of defective items in a large manufactured lot is unknown. The following prior distribution is used:

$$\pi(p) = \begin{cases} 2(1-p), & \text{for } 0 \le p \le 1 \\ 0, & \text{otherwise} \end{cases}.$$

What is the prior mean and variance for p?

2) $f(x) = \lambda e^{-\lambda x}$

$x=2 \rightarrow \lambda e^{-2\lambda} = L(\lambda | x=2)$

$x=2.5 \rightarrow \lambda e^{-2.5\lambda}$

$x=3 \rightarrow \lambda e^{-3\lambda}$

$\rightarrow L(\lambda | x) = \lambda^3 e^{-7.5\lambda}$

$\rightarrow \log L = l = 3 \cdot \log(\lambda) - 7.5\lambda$



Discussion Section 1
STAT 132 , Winter 2025

1. Suppose that the prior distribution of some parameter θ is a beta distribution for which the mean is $1/3$ and the variance is $1/45$. Determine the prior pdf of θ.

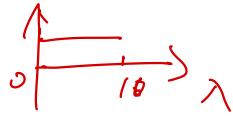
2. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. After randomly sampling 3 students, you found that they studied 2, 2.5, and 3 hours respectively. Write out the likelihood function for λ, and then draw a sketch of this function.

3. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. Your friend suggests a uniform distribution over the interval $[0, 10]$ for λ. Is this a reasonable prior distribution for this problem? Why or why not?

4. Suppose that the proportion p of defective items in a large manufactured lot is known to be either 0.1 or 0.2. Come up with a reasonable prior distribution for this problem, and then write out $\pi(p)$.

5. Suppose that the proportion p of defective items in a large manufactured lot is unknown. The following prior distribution is used:

$$\pi(p) = \begin{cases} 2(1-p), & \text{for } 0 \le p \le 1 \\ 0, & \text{otherwise} \end{cases}.$$



What is the prior mean and variance for p?

3. $f(x|\lambda) = \lambda e^{-\lambda x}$ $\quad \lambda \sim \text{Unif}[0, 10]$

$E(X) = \frac{1}{\lambda}$ $\quad \text{Var}(X) = \frac{1}{\lambda^2}$



Discussion Section 1
STAT 132 , Winter 2025

1. Suppose that the prior distribution of some parameter θ is a beta distribution for which the mean is $1/3$ and the variance is $1/45$. Determine the prior pdf of θ.

2. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. After randomly sampling 3 students, you found that they studied 2, 2.5, and 3 hours respectively. Write out the likelihood function for λ, and then draw a sketch of this function.

3. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. Your friend suggests a uniform distribution over the interval $[0, 10]$ for λ. Is this a reasonable prior distribution for this problem? Why or why not?

4. Suppose that the proportion p of defective items in a large manufactured lot is known to be either 0.1 or 0.2. Come up with a reasonable prior distribution for this problem, and then write out $\pi(p)$.

5. Suppose that the proportion p of defective items in a large manufactured lot is unknown. The following prior distribution is used:

$$\text{prior} \rightarrow \quad \pi(p) = \begin{cases} 2(1-p), & \text{for } 0 \le p \le 1 \\ 0, & \text{otherwise} \end{cases} .$$

What is the prior mean and variance for p?

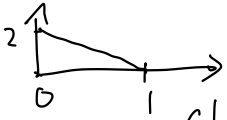
4. $\pi(p) = \begin{cases} 0.5 & p = 0.1 \\ 0.5 & p = 0.2 \end{cases}$

$2p$

$\frac{4-3}{6}$ $p^2 - \frac{2}{3}p^3 \Big|_0^1$

$\frac{2}{3}p^3 - \frac{1}{2}p^4 \Big|_0^1$

5. ① check pdf



$E(p) = \int_0^1 p \cdot \pi(p)\, dp = \int_0^1 2p(1-p)\, dp = \frac{1}{3}$

$\frac{1}{6} - \frac{1}{9} = \frac{1}{18}$ $= Var(p) = E(p^2) - [E(p)]^2$

$= \int_0^1 p^2 \pi(p)\, dp = \int_0^1 2p^2(1-p)\, dp = \frac{1}{6}$